

Money Growth and Inflation After 2008 and COVID

Motivation & Empirical Design

Motivated by Lucas (1996) and McCandless & Weber (1995), this memo summarizes my study that revisits the quantity theory of money to ask: does the money-inflation relationship look the same across horizons, samples, and regimes—especially post-2008 and post-COVID?

I constructed a 160-country annual panel (1991–2024) using World Bank WDI data. Following the approach of Lucas (1996) and McCandless & Weber (1995), I extend and focus on two main empirical designs: 1. **Long-run cross-country averages** ($n \geq 30$) and 2. **Short-run within-country pass-through** (Two-Way Fixed Effects) to capture the Money-Inflation relationship.

To stratify the sample and to prevent outliers from dominating the estimation, I construct a “clean” sample (excluding countries with any inflation year $> 40\%$) and also filter the 1/99 percentiles of the full sample. They are then to compare with the full sample results to understand how the relationship varies across regimes and over time. To note, my data cleaning process removes potential data issues, e.g., a spurious -680% M2 unit break in Sierra Leone. All estimates are descriptive associations rather than causal.

Main Results and Interpretation

The interpretation is multi-fold: horizon (long/short run) and regime-dependent (as noted by Borio et al., 2024).

1. The Long-Run Relationship Survives

Table 1 summarizes the main evidence. Clearly, money growth still organizes long-run inflation differences across countries. The full long-run country-average slope is 0.952, close to the one-for-one quantity-theory benchmark. In the **clean sample**, the long-run slope falls to 0.524 but remains positive. This is my first key result: the long-run relationship definitely holds even after 30 years, with the new data.

However, its magnitude is clearly regime-dependent: the full sample slope is close to one, but the clean sample slope is about half. This suggests the 1:1 ‘Long-Run Law’ is not universal; it is heavily propped up by hyperinflation regimes. In modern stable economies, the relationship attenuates by nearly half, challenging the classical proportionality assumption. That might imply a rather nonlinear relationship between money and inflation.

2. Short-Run Pass-Through is Weak and Non-Linear

The short-run result is interesting. In the full within-country fixed-effects specification, the slope is 0.855, but this estimate is heavily influenced by high-inflation regimes. When winsorizing the top and bottom 1% to mitigate these outliers, the slope drops to 0.422. In the clean sample, the same-year within-country slope is only 0.096, a result that remains robust even when applying Driscoll-Kraay standard errors to account for cross-sectional spatial correlation. This is one of the well-established facts: the long-run country-average relationship is strong, while annual within-country pass-through is weak. However, I am more interested in

the idea that, when extreme-inflation dynamics are removed, the annual relationship is even weaker; this implies there may be some nonlinear relationship that varies with the current inflation level (consistent with Borio et al., 2024).

Table 1: Core Result Map

Empirical question	Coef.	N	What it says
Long-run country averages: full sample	0.952	108	Near one-for-one
Clean sample	0.524	83	Still positive
Short-run country/year FE: full sample	0.855	4,749	High-inflation regimes dominate
Winsorized 1/99	0.422	4,749	Mitigates hyperinflation skew
Clean sample	0.096	3,638	Weak annual pass-through
Clean sample, 1991–2007	0.113	1,764	Positive pre-2008 slope
Clean sample, 2008–2019	-0.006	1,397	Near-zero post-2008 slope
Clean sample, 2020–2024	-0.014	477	Near-zero same-year COVID-era slope

Notes: Long-run rows use country averages with at least 30 observations. Short-run rows use country and year fixed effects. The clean sample excludes countries with any inflation year above 40 percent.

3. The Post-2008 Decoupling

After 2008, broad money growth is no longer a reliable stand-alone annual indicator of inflation in low- and moderate-inflation environments. In the clean sample, the annual within-country slope is 0.113 before 2008. It falls to -0.006 in 2008–2019 and remains near zero at -0.014 in 2020–2024. It implies that annual broad money growth has little contemporaneous association with inflation inside stable-inflation countries after 2008. The appendix figures show the same pattern visually: a strong between-country wedge, a post-2008 drop in the clean annual slope, and larger slopes in higher-prior-inflation regimes.

While a weak short-run link is expected in the literature due to noise, the key is it now implies that we could observe high money supply growth for a prolonged period until inflation finally rises. A cautious reading is that annual money growth is a noisy signal when inflation is low and policy regimes change. But it might also imply a loose decoupling of money growth and inflation in the post-2008 era, which could be due to changes in monetary policy frameworks, financial innovation, or structural changes in the economy that affect money demand. Decoupling does not mean the long-run relationship is dead (shown above), but it does suggest that the short-run pass-through has been slowed, although the QTM does not break down.

4. The COVID-Era Anomaly

COVID-era results are also interesting. The clean-sample short-run slope is near zero in 2020–2024: early money growth coincided with deflationary pressure from the recession. But the QTM will return with even bigger force: the 2022–2024 inflation spike, which is partly explained by the previous money growth. While the relationship is not strong and one-to-one, my estimate shows countries that printed more money on average during the pandemic tended to have higher average inflation in the COVID era (Figure 3). The distributed-lag estimates (Appendix B) show that pass-through accumulates over 2–3 years rather than dissipating. This implies that, while the same-year association is near zero, the effect arrives later rather than never — monetary expansion during a downturn may translate into inflationary pressure as the economy revives and money demand normalises.

5. Inflation Targeting and Expectations

(exploratory; endogenous adoption, not a causal treatment effect)

Another extension is that inflation targeting compresses inflation dispersion, potentially weakening the money-inflation relationship. My interpretation is that inflation targeting sets a fixed rule that shapes expectations and behavior. Even if fiscal authorities print extra money, or there are changes in velocity or the reserve multiplier leading to more deposits created, the central bank is expected to independently defend the inflation target and adjust the policy rate to offset the inflationary pressure. Thus, it might then reduce the sensitivity of inflation to money growth. The key is, when estimating a staggered Difference-in-Differences event study, inflation-targeting adopters show lower post-adoption money-inflation slopes (a dynamic consistent with expectations anchoring, as in Teles et al., 2016), but adoption timing is endogenous, so this is descriptive heterogeneity rather than a treatment effect.

Exploratory Appendices

Rest of the appendix checks are meant to be exploratory. Distributed-lag estimates agree with the above that associations accumulate over multiple years, helping reconcile weak same-year pass-through with stronger long-run relationships.

LP-IV (weak identification, $F=5.96$; directional only, not a headline claim.)

Out of curiosity, I constructed a Bartik-style shift-share instrument based on external lending rates. The LP-IV results are positive, but the primary first-stage F-statistic is 5.96, below the conventional strong-IV threshold; lagged M2 is mechanically stronger but not credible as an excluded instrument.

The U.S. Lucas-style (1980) M2 appendix tells a similar story: the low-frequency M2-inflation relationship is strong over the long sample, but flatter after 2008. The less positive slope, in line with the above findings, implies a loosening of money growth and inflation in the post-2008 era. Use of M2 instead of M1 due to change in M1 definition in 2020.

Conclusion

The long-run cross-country money-inflation relationship remains strong through 2024: a slope near one in the full sample and 0.52 in the low-inflation regime, consistent with the quantity theory.

The short-run within-country association is a different story, where the clean-sample TWFE slope was 0.11 before 2008 and is statistically indistinguishable from zero afterward. Pass-through did not vanish; it slowed. Distributed-lag estimates show the association accumulates over two to three years, reconciling the near-zero annual signal with the durable long-run relationship.

Appendix: Figure and Table Pack

A. Core Visual Evidence

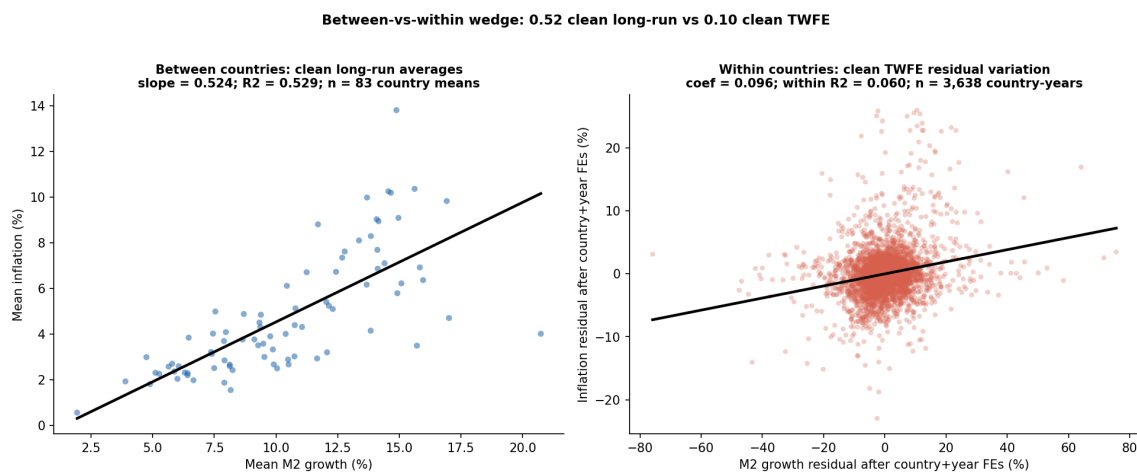


Figure 1: The core contrast: Long-run between-country association is strong; short-run within-country pass-through is weak.



Descriptive association; not causal. Separate period TWFE with country+year FEs, clustered by country. The 2020-2024 bin has only five years, so read it as a noisy COVID-era check.

Figure 2: Post-2008 Decoupling: Clean annual within-country slope collapses to zero.

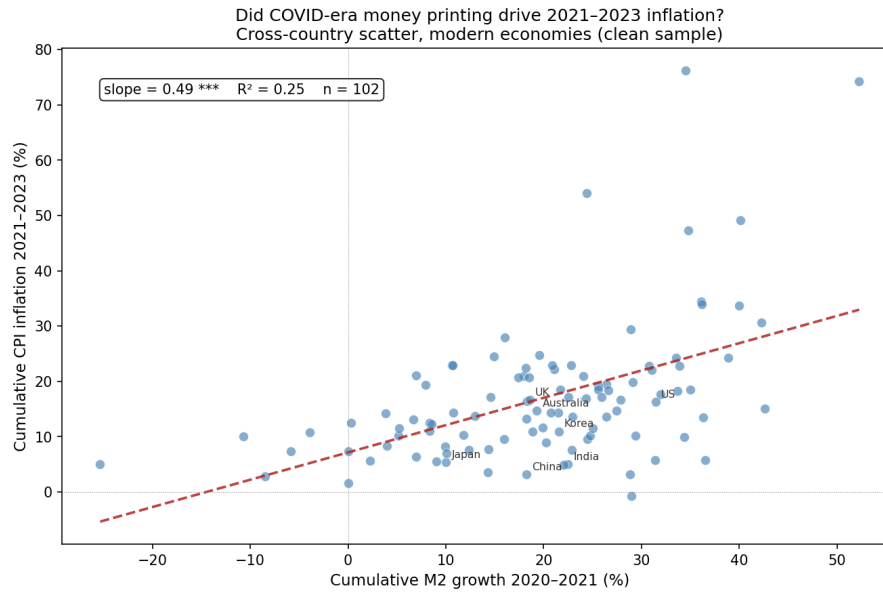


Figure 3: COVID Stress Evidence: Countries with higher money growth during 2020–2021 tended to experience higher average inflation in the post-COVID era (descriptive association).

Table 2: Short-Run Robustness and Outlier Diagnostics

Specification	Coef.	SE	N
Full sample, clustered SE	0.8549	0.0757	4,749
Clean sample, clustered SE	0.0961	0.0132	3,638
Clean sample, excluding 2020–2021	0.0990	0.0139	3,430
Clean sample, Driscoll–Kraay SE	0.0961	0.0424	3,638
Winsorized 1/99 sample	0.4222	0.0481	4,749
Exclude $ M2 \text{ growth} > 1.0$	0.3278	0.0585	4,717
Exclude inflation > 0.40	0.1140	0.0146	4,621

Notes: Same country/year FE estimand. The clean result holds under Driscoll–Kraay spatial standard errors and leaving out COVID years.

B. Dynamics and Regime Heterogeneity

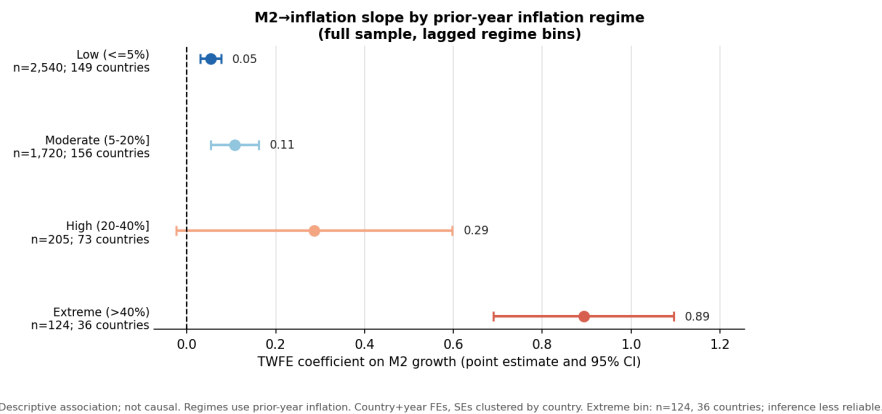


Figure 4: Inflation-Regime Split: M2-Inflation slope rises sharply with prior-year inflation level — near zero in low-inflation environments, close to one in extreme regimes.

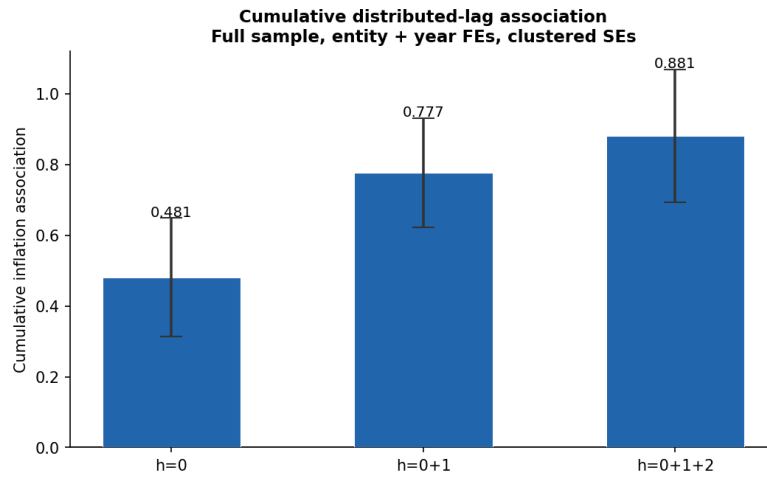


Figure 5: Distributed Lags: Pass-through accumulates over multiple years, approaching the long-run slope.

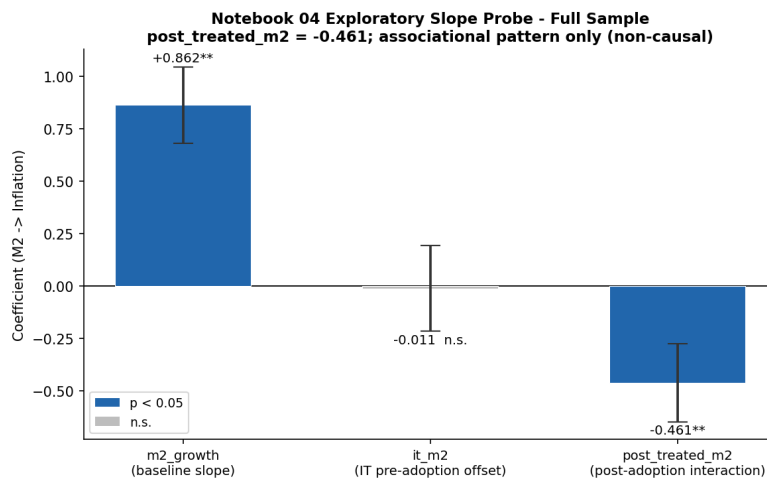


Figure 6: Inflation Targeting: Adopters show a lower post-adoption M2-Inflation slope; associational heterogeneity only.

C. U.S. and Identification Appendices

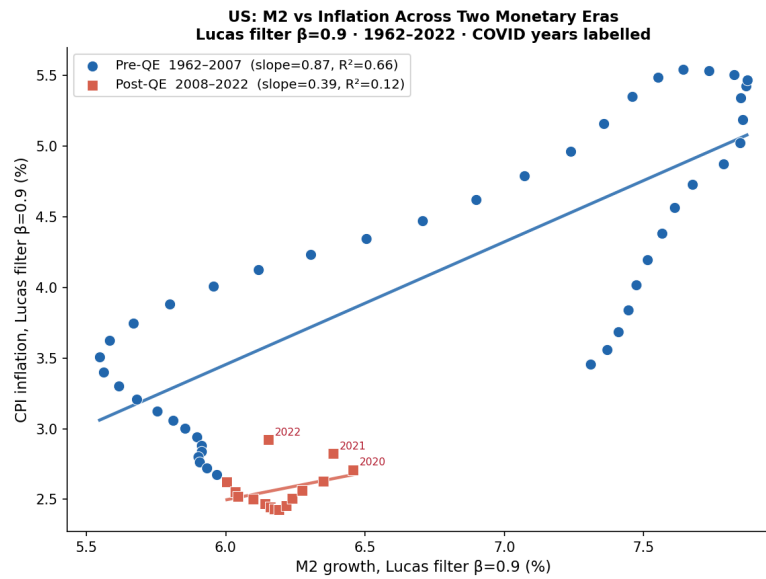


Figure 7: U.S. M2 (Lucas Filtered): Strong historical relationship flattens severely post-2008.

Table 3: Why IV Results Stay Appendix-Only

Instrument	First-stage F	Fixed N	Interpretation
External-rate exposure (Bartik)	5.96	3,382	Relevant but weak ($F < 10$); directional only.
Lagged M2 growth	16.29	3,382	Strong first stage, but exclusion restriction is not credible.

Notes: Weak instruments prohibit causal headline claims. OLS remains the primary descriptive benchmark.